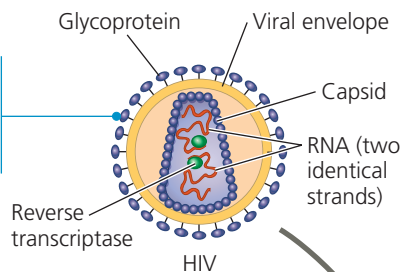


1 The envelope glycoproteins enable the virus to bind to specific receptors (not shown) on certain white blood cells.



▼ Figure 19.9 The replicative cycle of HIV, the retrovirus that causes AIDS. The photos on the left (artificially colored TEMs) show HIV entering and leaving a human white blood cell. See Figure 7.8 for the cell-surface proteins that act as receptors for HIV. Note in step **6** that DNA synthesized from the viral RNA genome is integrated as a provirus into the host cell chromosomal DNA, a characteristic unique to retroviruses.

2 The virus fuses with the host cell's plasma membrane.

3 The capsid proteins are removed, releasing the viral proteins, RNA, and reverse transcriptase.

4 Reverse transcriptase catalyzes the synthesis of a DNA strand, using the viral RNA as a template and then hydrolyzing it.

5 Reverse transcriptase catalyzes the synthesis of a second DNA strand complementary to the first.

6 The double-stranded DNA is incorporated as a provirus into the cell's DNA.

7 Proviral genes are transcribed into RNA molecules, which serve as genomes for progeny viruses and as mRNAs for translation into viral protein.

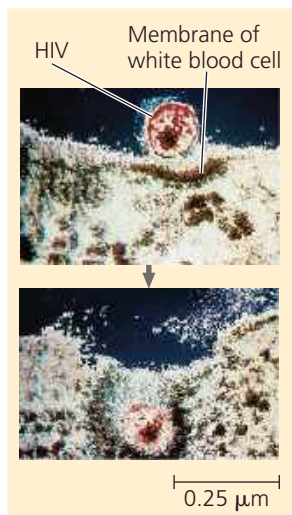
8 Viral envelope glycoproteins are made in the ER.

9 Viral capsid proteins and reverse transcriptase are made in the cytosol.

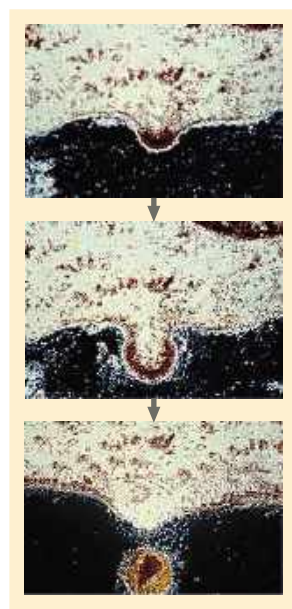
10 Vesicles transport the glycoproteins to the cell's plasma membrane.

11 Capsids are assembled around viral genomes and reverse transcriptase molecules.

12 New viruses, with viral envelope glycoproteins, bud from the host cell.



HIV entering a cell



New HIV leaving a cell

MAKE CONNECTIONS Describe what is known about binding of HIV to immune system cells. (See Figure 7.8.) How was this discovered?

➔ **Mastering Biology** HHMI Animation: HIV Replicative Cycle
Animation: Retrovirus (HIV) Replicative Cycle

